Yifan Kang

189 New Vassar St, Cambridge, MA 02139 yifank@mit.edu yifank.github.io

Education

Massachusetts Institute of Technology Cambridge, MA
B.S. in Mathematics and Computer Science 2024 – Present

- GPA: 5.0/5.0 **Selected Coursework:** Discrete Probability & Stochastic Processes (18.619), Probability (6.7700), Advanced Algorithms (6.5210), Deep Learning (6.7960), Robotics: Science & Systems (6.4200).

Research Experience

Research Intern — 2077AI Jun 2026 – Present

- *Encore* — an agentic few-shot imitation system: a coding agent turns 3 demonstrations into an executable policy program and self-repairs it from execution traces (**Oral, NeurIPS 2026 Robot Learning Workshop**); deployed on a real-robot ALOHA bimanual platform.

Student Researcher — MIT, with Prof. Leslie Pack Kaelbling Jun 2025 – Present

- *KeyFlow: Agentic Policy Adaptation with Episodic Memory* — a coding agent adapts a manipulation policy by rewriting how a few demonstrations are represented and retrieved, **62.6%** mean held-out success across 13 RoboCasa tasks vs. 47.8% for a VLA fine-tuned on 10× more demonstrations (Under review).

Student Researcher — MIT, with Prof. Antonio Torralba Jun 2025 – Aug 2025

- *Doodle Agent: Exploring Freeform Visual Generation with Multimodal LLMs* — co-first author (**ICCV 2025 AI4VA Workshop** Accepted).

Interpreting Diffusion Language Models Over Denoising Time 2026
Sparse autoencoders and cross-layer transcoders for diffusion LLMs (**CoLM 2026 Workshop on Non-Autoregressive LMs** Accepted).

Research Science Institute (RSI) Scholar Jun 2023 – Aug 2023

- *CNOT-Optimal Circuit Synthesis* — selected as a **top 5 paper at RSI** out of 100 candidates.

Selective Programs

Explore HRT — Hudson River Trading, Algorithm Development (Quantitative Research) track Apr 2026
Implemented a **limit order book** and an **arbitrage strategy** from scratch; **ranked 4th of 15** by PnL.
Citadel Securities Quant Invitational — one of **24 students** invited nationwide Jan 2026

Projects

Sparkie — AI meeting agent Sep 2026
Built at **HackMIT**: joins a Zoom call as a participant and does the work in the background — a voice agent decides whether to answer or delegate to a coding worker, so the meeting never blocks.

Autonomous Racecar Feb 2026 – May 2026
Built a full autonomy stack on a **ROS2 racecar** (LiDAR, camera, odometry): particle-filter localization, path planning, and pure-pursuit control; final challenge combined high-speed lane keeping with **YOLO**-based autonomous parking and traffic-light handling.

Awards & Honors

MIT CSAIL Agentic AI Hackathon — Winner (Jane Street Award) 2026
William Lowell Putnam Mathematical Competition — Rank 254 2024
United States of America Mathematical Olympiad (USAMO) — Honorable Mention 2023
United States of America Computing Olympiad (USACO) — Finalist (Top 26) 2022
United States of America Physics Olympiad (USAPhO) — Bronze Medalist 2022

Leadership

Vice President — MIT Informatics Tournament (MITIT)
Help lead an international competitive programming tournament whose 2025 Winter event reached **2,447 contestants across 91 countries**, including **1,073 participants with no prior experience**.

Technical Skills

Programming: Python (PyTorch), C++, Rust, Java **Languages:** English, Chinese (Bilingual), Japanese (JLPT N1)